

**T104-BASIC ELECTRICAL AND ELECTRONICS ENGINEERING –
PART B (ELECTRONICS)
UNIT – IV ELECTRONIC CIRCUITS**

V-I Characteristics of diode - Half-wave rectifier and Full-wave rectifier – with and without capacitor filter - Transistor - Construction & working - Input and output characteristics of CB and CE configuration - Transistor as an Amplifier - Principle and working of Hartley oscillator and RC phase shift oscillator - Construction and working of JFET & MOSFET.

2 MARKS

1. What is known as the breakdown voltage of PN Diode?(Jan-2016)

The maximum reverse bias voltage that can be applied to a p-n diode is limited by breakdown. Breakdown is characterized by the rapid increase of the current under reverse bias. The corresponding applied voltage is referred to as the breakdown voltage.

2. Which are the types of Transistor circuit configurations?(Jan-2016)

There are three types of configuration for the operation of a transistor

- Common base (CB) configuration
- Common emitter (CE) configuration
- Common collector (CC) configuration

3. What is doping.(May-2016, May-2018)

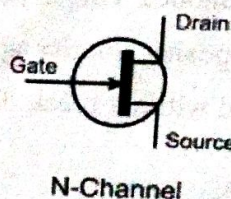
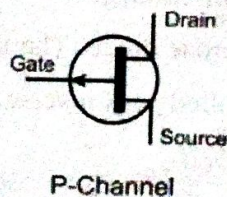
Doping is the process by which conductivity of the semiconductor is increased. Doping means adding an impurity into a pure form of semiconductor to improve its electrical conductivity.

4. Define Rectification Efficiency. (Nov-2017)

Rectification Efficiency is quantitative measurement of rectifier efficiency. It is ratio of DC power output to the AC power input of the rectifier. It is usually denoted by η and expressed in percentage.

$$\text{Rectification Efficiency, } \eta = \frac{\text{DC Power Output}}{\text{AC Power Input}}$$

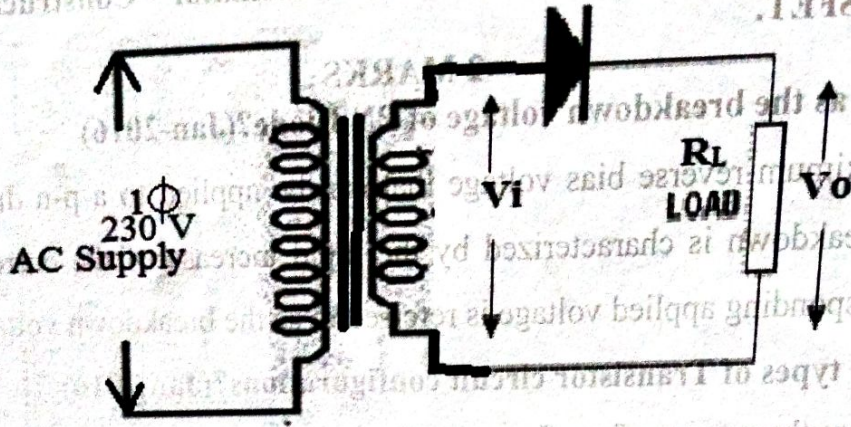
5. Draw the symbol of two types of JFET.(Nov-2017)



6. How many diodes are there in JFET? (May-2017)

A JFET is similar to two diodes. The gate and the source form one of the diodes, and the drain form the other diode. These two diodes are usually referred as the gate-source diode and the gate-drain diode.

7. Draw the circuit diagram of half-wave rectifier. (Dec-2016)



8. Compare JFET and MOSFET. (Dec-2016)

JFET	MOSFET
Operated in Depletion region	Operated in Depletion region and Enhancement region
High input Impedance	Very High Input Impedance
Gate is not insulated from Channel	Gate is insulated from Channel by a layer of SiO_2
Channel exists permanently	Channel exists permanently in depletion type but not in enhancement type
Difficult to fabricate than MOSFET	Easier to fabricate
Drain resistance is high	Drain resistance is less
Gate is formed as a diode	Gate is formed as a Capacitor

Define peak inverse voltage in diode. (Nov-2018)

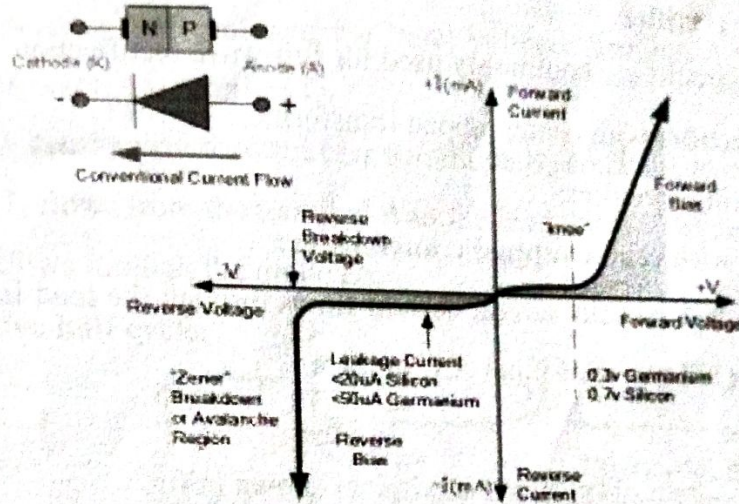
When diode is forward biased, some voltage is necessary to overcome the barrier potential, to make diode conduct. This is called cut-in-voltage.

In reverse bias opposite polarity appears across diode. The maximum diode voltage which diode can withstand without break down is called peak inverse voltage.

10. Why CE configuration is used in amplifier circuits.(Nov-2018)

Common emitter is the most basic configuration for amplifier circuits. It also provides the maximum transconductance or voltage gain for a given load. The common emitter configuration has the highest power gain combined with medium voltage and current gain.

11. Draw the V-I Characteristics of diode.(Sept-2020)



12. What is a Hartley Oscillator? (Sept-2020).

The Hartley oscillator is that the tuned circuit consists of a single capacitor in parallel with two inductors in series (or a single tapped inductor), and the feedback signal needed for oscillation is taken from the center connection of the two inductors.

11 MARKS

1. Draw the circuit diagram of a full wave rectifier network supplying a resistive load. Explain operation of the network relevant waveforms. (Apr-2014, Jan-2016, May-2016, May-2018, Sept-2020)

Full-Wave Rectifier (FWR)

Definition: In full wave rectifier circuit current flows through the load in the same direction for both half-cycles (full wave) of input a.c voltage.

Types of full wave rectifier:

The following two circuits are commonly used for full-wave rectification.

- Full wave rectifier with center tapped transformer.
- Full wave bridge rectifier.

Full wave rectifier with center tapped transformer:

Definition: In full wave rectifier circuit current flows through the load in the same direction for both half-cycles (full wave) of input A.C voltage.

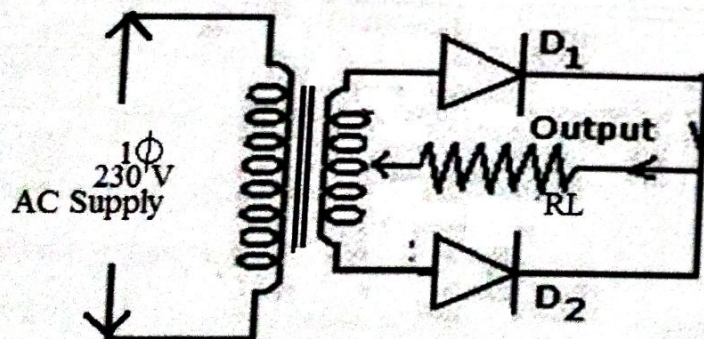
Circuit diagram:

The circuit diagram of full-wave rectifier is shown in fig.

This circuit contains

- Centre tapped transformer
- Two Diodes (D_1 & D_2)
- Load Resistor

Two diode D_1 and D_2 are used as the switching elements, one diode acts for half cycle and second diode for another half cycle.

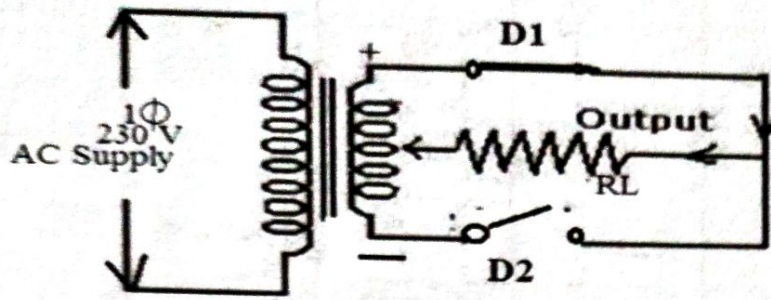


FWR with center tapped transformer

Operation of the circuit:

In A.C supply voltage, there are two half cycle, one is positive half cycle the other is negative half cycle.

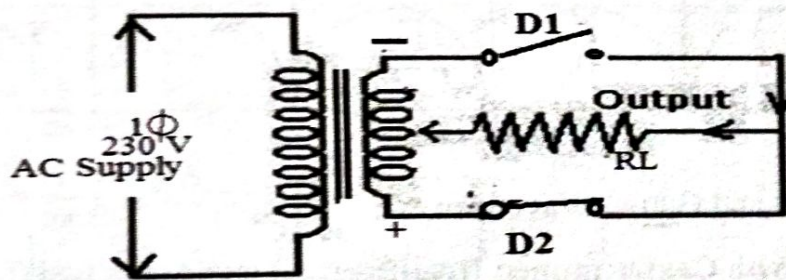
Duration of Positive half cycle:



During positive half cycle

- During the positive half cycles of the input voltage, terminal A is positive and B is negative with respect to terminal O.
- The diode D_1 conducts in forward bias and diode D_2 conducts in reverse bias.
- The current I_1 flows from the terminal A to the load through diode D_1 .
- No current flows through the diode D_2 .

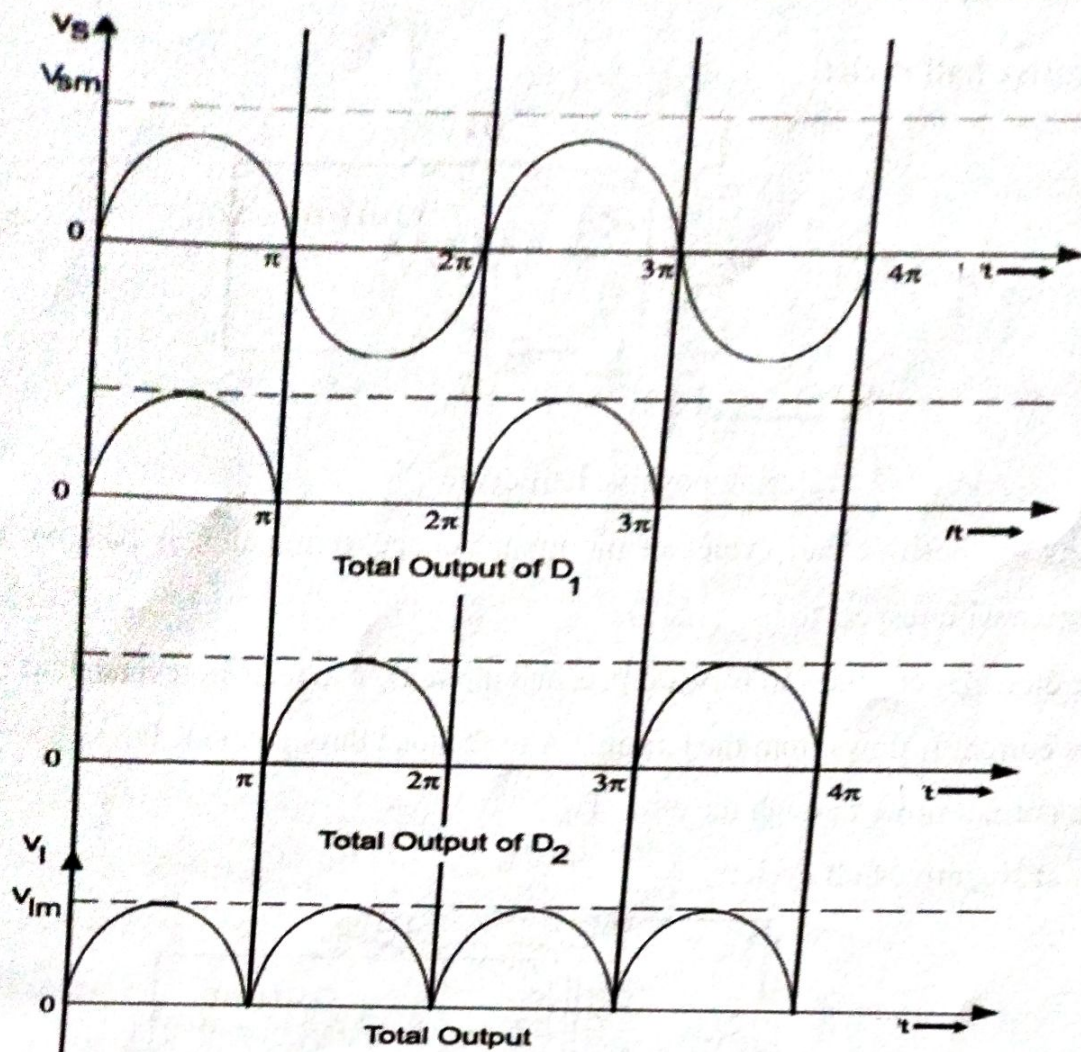
Duration of Negative half cycle:



During Negative Half cycle

- During the negative half cycles of the input voltage, terminal B is positive and A is negative with respect to terminal O.
- The diode D_2 conducts in forward bias and diode D_1 conducts in reverse bias.
- The current I_2 flows from the terminal B to the load through diode D_2 .
- No current flows through the diode D_1 .

Input and Output Waveforms:



Input and Output Waveforms

Analysis of FWR with Center tapped Rectifier:**D.C output Current:**

Input signal a.c voltage \$= V_m \sin \theta\$

Load resistance \$= R_L\$

Input signal a.c current \$= I_m \sin \theta\$

Hence the output d.c. current

$$I_{dc} = \frac{1}{2\pi} \int_0^{2\pi} i d\theta$$

$$= \frac{1}{2\pi} \int_0^{\pi} i d\theta + \frac{1}{2\pi} \int_{\pi}^{2\pi} i d\theta$$

The variation between \$0\$ and \$\pi\$ is equal to \$\pi\$ and \$2\pi\$.

$$I_{dc} = \frac{1}{2\pi} \int_0^{\pi} i d\theta + \frac{1}{2\pi} \int_0^{\pi} i d\theta$$

$$= \frac{1}{2\pi} \times 2 \left[\int_0^{\pi} i d\theta \right]$$

$$= \frac{1}{\pi} \int_0^{\pi} I_m \sin \theta d\theta$$

$$\begin{aligned}
 &= \frac{I_m}{\pi} [\cos \theta] \\
 &= \frac{I_m}{\pi} [-(\cos \pi - \cos 0)] \\
 &= \frac{I_m}{\pi} [-(-1 - 1)] \\
 &= \frac{2I_m}{\pi}
 \end{aligned}$$

$$I_{dc} = \frac{2I_m}{\pi}$$

D.C Voltage:

$$\begin{aligned}
 \text{D.C Voltage } V_{dc} &= I_{dc} R_L & V &= IR \\
 &= \frac{2I_m R_L}{\pi} & V_m &= I_m R_L \\
 V_{dc} &= \frac{2V_m}{\pi}
 \end{aligned}$$

A.C INPUT CURRENT:

$$\begin{aligned}
 I_{rms}^2 &= \frac{1}{2\pi} \int_0^{2\pi} i^2 d\theta \\
 &= \frac{1}{2\pi} \int_0^{\pi} i^2 d\theta + \frac{1}{2\pi} \int_{\pi}^{2\pi} i^2 d\theta
 \end{aligned}$$

The variation between 0 and π is equal to π and 2π .

$$\begin{aligned}
 I_{rms}^2 &= \frac{1}{2\pi} \int_0^{\pi} i^2 d\theta + \frac{1}{2\pi} \int_0^{\pi} i^2 d\theta \\
 &= \frac{1}{2\pi} \times 2 \int_0^{\pi} i^2 d\theta \\
 &= \frac{1}{\pi} \int_0^{\pi} (I_m \sin \theta)^2 d\theta \\
 &= \frac{I_m^2}{\pi} \int_0^{\pi} \left[\frac{1 - \cos 2\theta}{2} \right] d\theta \\
 &= \frac{I_m^2}{2\pi} \left[\int d\theta - \int \cos 2\theta d\theta \right] \\
 &= \frac{I_m^2}{2\pi} \left[\theta - \frac{\sin 2\theta}{2} \right] \\
 &= \frac{I_m^2}{2\pi} \left[\left(\pi - \frac{\sin 2\pi}{2} \right) - \left(0 - \frac{\sin 2 \times 0}{2} \right) \right] \\
 &= \frac{I_m^2}{2\pi} [\pi - 0]
 \end{aligned}$$

$$I_{rms}^2 = \frac{I_m^2}{2}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

RMS Load Voltage:

$$V_{rms} = I_{rms} R_L$$

$$= \frac{I_m R_L}{2}$$

$$V_m = I_m R_L$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$\text{D.C output power} = I_{dc} R_L$$

$$P = I^2 R$$

$$= \left(\frac{2I_m}{\pi} \right)^2 R_L$$

$$P_{dc} = \frac{4I_m^2}{\pi^2} R_L$$

$$\text{A.C input power} = I_{rms}^2 (r_f + R_L)$$

$$= \frac{I_m^2}{2} (r_f + R_L)$$

$$P_{ac} = \frac{I_m^2}{2} (r_f + R_L)$$

Rectifier Efficiency:

$$\% \eta = \frac{P_{dc}}{P_{ac}} \times 100\%$$

$$= \frac{\left(\frac{2I_m}{\pi} \right)^2 R_L}{\frac{I_m^2}{2} (r_f + R_L)} \times 100\%$$

$$= \frac{4 \times I_m^2 \times R_L \times 2}{\pi^2 \times I_m^2 \times (r_f + R_L)} \times 100\%$$

$$= \frac{8}{\pi^2} \left(\frac{R_L}{r_f + R_L} \right) \%$$

$$= 81.2 \left(\frac{R_L}{r_f + R_L} \right) \%$$

If $r_f \ll R_L$

$$\eta = 81.2\%$$

Ripple Factor:

$$\text{Ripple Factor} = \sqrt{\left(\frac{I_{rms}}{I_{dc}} \right)^2 - 1}$$

$$= \sqrt{\left(\frac{I_m / \sqrt{2}}{2I_m / \pi} \right)^2 - 1}$$

$$= \sqrt{\left(\frac{\pi}{2\sqrt{2}} \right)^2 - 1}$$

$$= 0.483$$

$$\text{Ripple Factor} = 0.483$$

Ripple Frequency:

The frequency of the pulsating load voltage is known as ripple frequency. It is equal to the frequency of supply voltage in the half-wave rectifier.

$$f = 2f_s$$

Advantages:

1. The output voltage and transformer efficiency are higher.
2. Low ripple factor
3. High transformer Utilization factor.

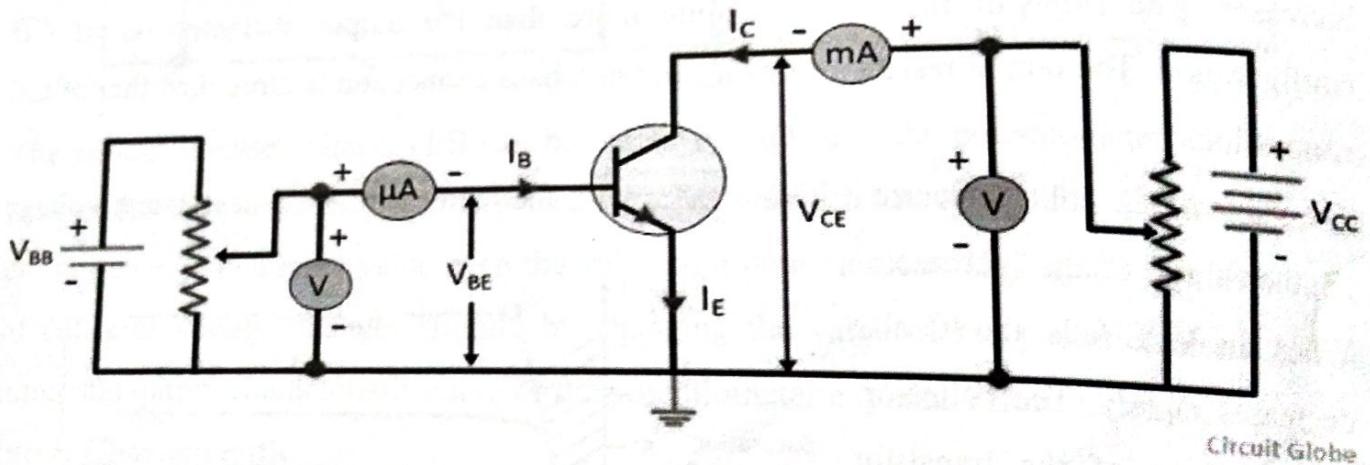
Disadvantages:

1. Usage of additional diode and bulky transformer is needed, and hence increase in cost.
2. The peak inverse voltage of diode is high.

3. With a neat circuit diagram, explain the operation of CE Configuration of Transistor. (Nov-2016, May-2017)

The characteristic of the common emitter transistor circuit is shown in the figure below. The base to emitter voltage varies by adjusting the potentiometer R1. And the collector to emitter voltage varied by adjusting the potentiometer R2. For the various setting, the current and voltage are taken from the milliammeters and voltmeter. On the basis of these readings, the input and output curve plotted on the curve.

Characteristic-Curve

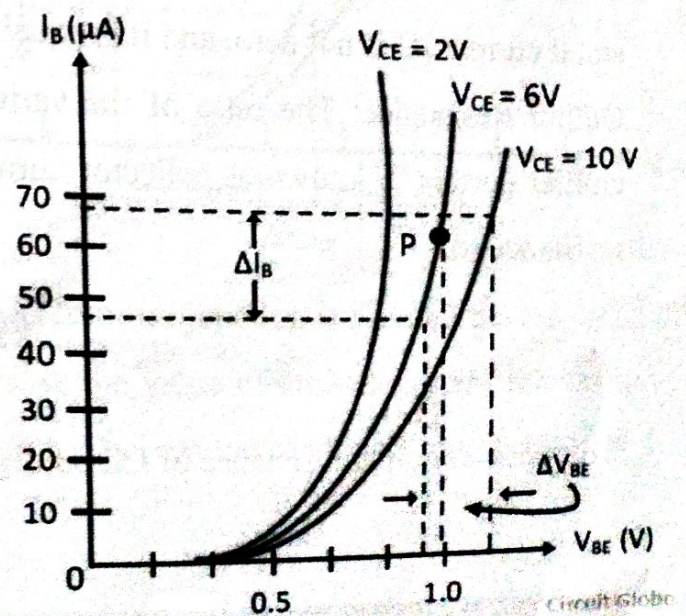


Input Characteristic Curve

The curve plotted between base current I_B and the base-emitter voltage V_{BE} is called Input characteristics curve. For drawing the input characteristic the reading of base currents is taken through the ammeter on emitter voltage V_{BE} at constant collector-emitter current. The curve for different value of collector-base current is shown in the figure below.

Input-Characteristic

The curve for common base configuration is similar to a forward diode characteristic. The base current I_B increases with the increases in the emitter-base voltage V_{BE} . Thus the input resistance of the CE configuration is comparatively higher that of CB configuration. The effect of CE does not cause large deviation on the curves, and hence the effect of a change in V_{CE} on the input characteristic is ignored.



Input Resistance: The ratio of change in base-emitter voltage V_{BE} to the change in current ΔI_B at constant collector-emitter voltage V_{CE} is known as input resistance, i.e.,

$$r_i = \frac{\Delta V_{BE}}{\Delta I_B} \text{ at constant } V_{CE}$$

Output Characteristic

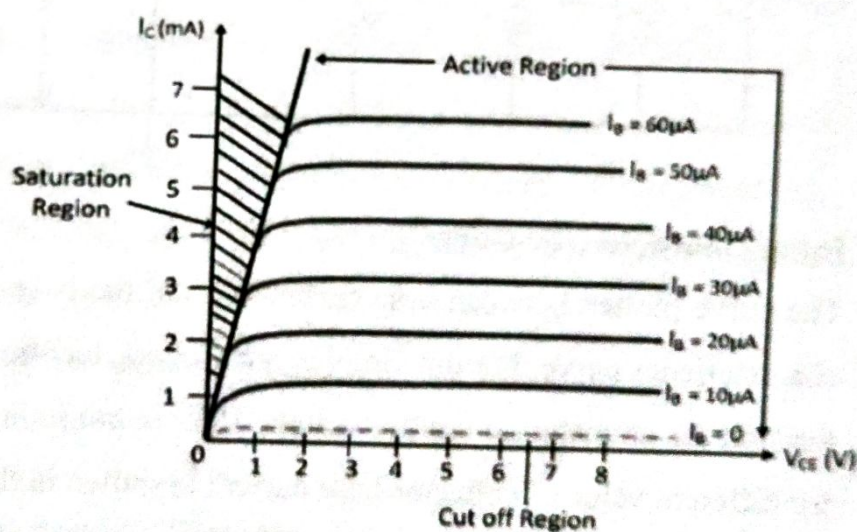
In CE configuration the curve draws between collector current I_C and collector-emitter voltage V_{CE} at a constant base current I_B is called output characteristic. The characteristic curve for the typical NPN transistor in CE configuration is shown in the figure below.

Output-Characteristic-Curve

In the active region, the collector current increases slightly as collector-emitter V_{CE} current increases. The slope of the curve is quite more than the output characteristic of CB configuration. The output resistance of the common base connection is more than that of CE connection.

The value of the collector current I_C increases with the increase in V_{CE} at constant voltage I_B , the value β of also increases.

When the V_{CE} falls, the I_C also decreases rapidly. The collector-base junction of the transistor always in forward bias and work saturate. In the saturation region, the collector current becomes independent and free from the input current I_B



Output Characteristic Curve

In the active region $I_C = \beta I_B$, a

small current I_C is not zero, and it is equal to reverse leakage current I_{CEO} .

Output Resistance: The ratio of the variation in collector-emitter voltage to the collector-emitter current is known at collector currents at a constant base current I_B is called output resistance r_o .

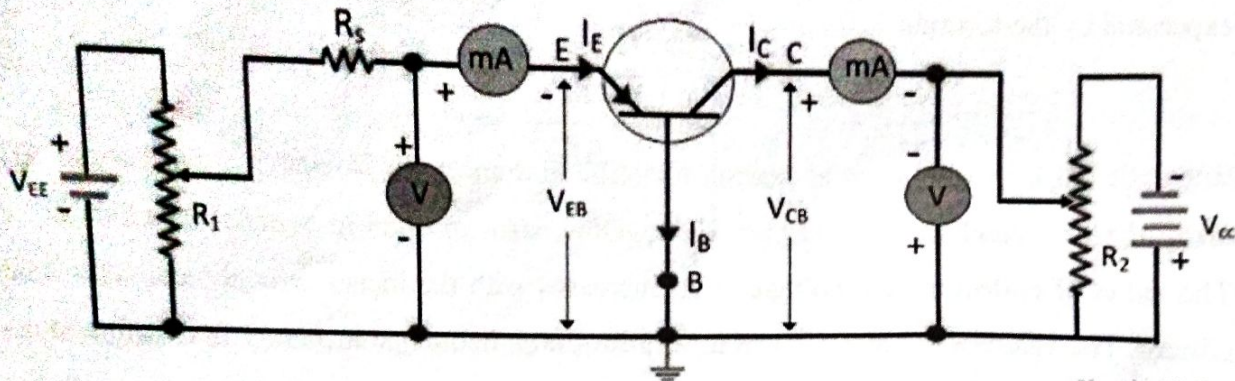
$$r_o = \frac{\Delta V_{CE}}{\Delta I_C} \text{ at constant } I_B$$

The value of output resistance of CE configuration is more than that of CB

4. Explain the working of BJT in CB Configuration with suitable diagrams and characteristic curves.(Nov-2017, Sept-2020)

The characteristic diagram of determining the common base characteristic is shown in the figure below.

Common-Base-Characteristic

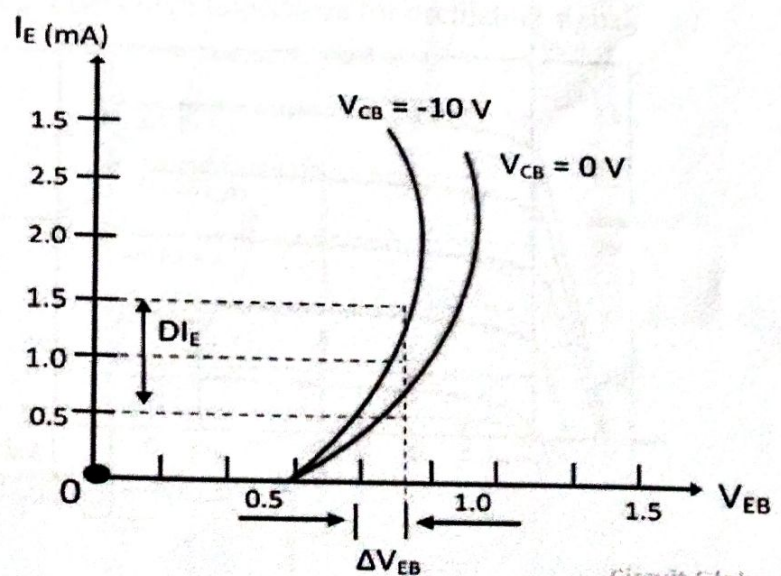


The emitter to base voltage V_{EB} can be varied by adjusting the potentiometer R_1 . A series resistor R_S is inserted in the emitter circuit to limit the emitter current I_E . The value of the emitter change to a large value even the value of a potentiometer slightly change. The value of collector voltage changes slightly by changing the value of the potentiometer R_2 . The input and output characteristic curve of the potentiometer explains below in details.

Input Characteristic

The curve plotted between emitter current I_E and the emitter-base voltage V_{EB} at constant collector base voltage V_{CB} is called input characteristic curve. The input characteristic curve is shown in the figure below.

The following points are taken into consideration from the characteristic curve.



For a specific value of V_{CB} , the curve is a diode characteristic in the forward region. The PN emitter junction is forward biased. When the value of the voltage base current increases the value of emitter current increases slightly. The junction behaves like a better diode. The emitter and collector current is independent of the collector base voltage V_{CB} .

The emitter current I_E increases with the small increase in emitter-base voltage V_{EB} . It shows that input resistance is small.

Input Resistance

The ratio of change in emitter-base voltage to the resulting change in emitter current at constant collector base voltage V_{CB} is known as input resistance. The input resistance is expressed by the formula

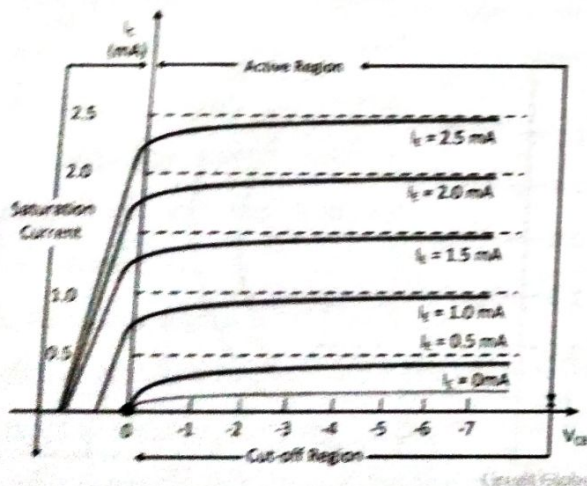
$$r_i = \frac{\Delta V_{EB}}{\Delta I_E}$$

Output-Characteristic

The value of collector base voltage V_{CB} increases with the increases in the collector-base current. The value of input resistance is very low, and their value may vary from a few ohms to 10 ohms.

Output Characteristic Curve

In common base configuration, the curve plotted between the collector current and collector base voltage V_{CB} at constant emitter current I_E is called output characteristic. The CB configuration of PNP transistor is shown in the figure below. The following points from the characteristic curve are taken into consideration.



The active region of the collector-base junction is reverse biased, the collector current I_C is almost equal to the emitter current I_E . The transistor is always operated in this region.

The curve of the active regions is almost flat. The large changes in V_{CB} produce only a tiny change in I_C . The circuit has very high output resistance

ro.

When V_{CB} is positive, the collector-base junction is forward bias and the collector current decrease suddenly. This is the saturation state in which the collector current does not depend on the emitter current.

When the emitter current is zero, the collector current is not zero. The current which flows through the circuit is the reverse leakage current, i.e., I_{CBO} . The current is temperature

depends and its value range from 0.1 to 1.0 μA for silicon transistor and 2 to 5 μA for germanium transistor.

Output Resistance

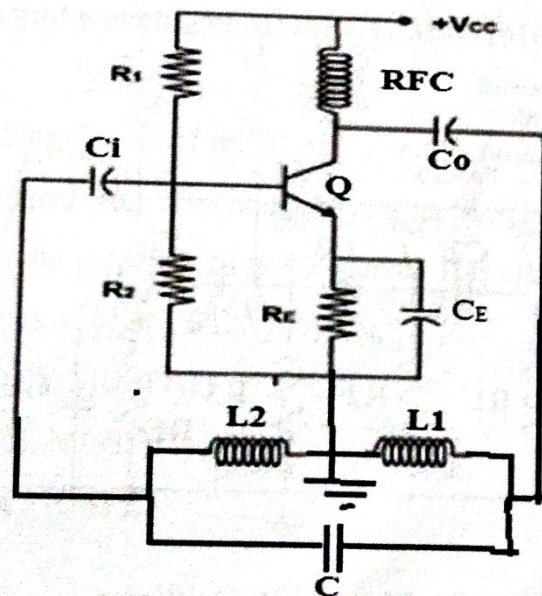
The ratio of change in collector-base voltage to the change in collector current at constant emitter current I_E is known as output resistance.

$$r_o = \frac{\Delta V_{CB}}{\Delta I_C}$$

The output characteristic of the change in collector current is very little with the change in VCB. With the change in collector-base voltage. The output resistance is very high of the order of several kilometres.

5. With a neat circuit diagram, explain the construction and operation of Hartley oscillator. (Jan-2016, Apr-2014, May-2017, Nov-2017)

The Circuit Diagram of a Hartley oscillator is shown in fig., it is a LC oscillator. This oscillator contains a CE amplifier, feedback network, and a tank circuit made up of L_1 , L_2 and C. The resistors R_1 and R_2 provide necessary bias to the amplifier. The capacitors C_i and C_o are used to block the DC components. The capacitor C_E is a bypass capacitor. The resistor R_E provides negative feedback to the amplifier to improve its stability. The RF choke (RFC) provides a path for collector bias current but offers high impedance for oscillating signal.



Hartley Oscillator

Principle of operation:

When the supply is turned ON, the capacitor 'C' is charged. When this capacitor is fully charged, it discharges through the coils L_1 and L_2 , setting up an oscillation. The output

voltage of the amplifier appears across L_1 and the feedback voltage appears across L_2 . The voltage across L_2 is 180° out of phase with the output voltage. it is the feedback signal. A phase shift of another 180° is produced by CE amplifier. Hence the total phase shift between input and output is $180^\circ + 180^\circ = 360^\circ$. This results in positive feedback which makes the oscillation as continuous undamped.

The frequency of the oscillation is given by,

$$F = \frac{1}{2\pi\sqrt{LC}}$$

Where, $L = L_1 + L_2$

It is used in local oscillators of radio receiver and audio oscillator circuits.

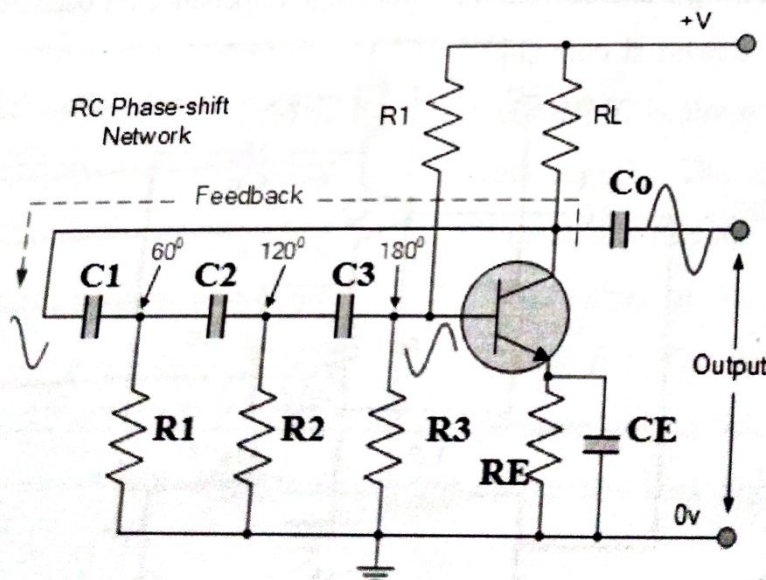
6. Explain the working principle of RC phase oscillator with a neat circuit diagram.(Nov-2016)

RC Phase Shift Oscillator:

The LC oscillator has two general drawbacks,

- It cannot be used for very low frequencies, because the coil becomes too much bulky, expensive and noisy.
- The frequency stability and waveform are very low.

The RC oscillator produce good frequency stability signal and also operate at very Low frequencies.



RC Phase Shift Oscillator

The circuit diagram of RC phase shift oscillator is shown in fig. The oscillator consists of three stages of RC networks (R_1C_1 , R_2C_2 and R_3C_3). The resistor R_5 provides bias, and R_E with C_E provides stabilization. The frequency of the oscillation is given by,

$$F = \frac{1}{2\pi R C \sqrt{6}}$$

Where, $R = R_1 = R_2 = R_3$ and $C = C_1 = C_2 = C_3$

Principle of Operation

When the supply is switched ON, the random variations of base current caused by noise variations in the transistor and voltage variations in the power source produce oscillation. This variation is amplified by the CE amplifier.

The feedback network consists of three stages of RC networks. The three stages are identical. The feedback section provides 180° phase shift because each RC network provides 60° phase shift ($3 \times 60^\circ = 180^\circ$). The CE amplifier provides another 180° phase shift. Hence the total phase shift is 360° , which provides positive feedback. Therefore, continuous undamped oscillation is produced.

Advantages:

- i.) Does not require transformers and inductors.
- ii.) It can produce very low frequency signals.
- iii.) The circuit provides good frequency stability.

Disadvantage:

- i.) It is difficult to start oscillation.
- ii.) It gives low power output.

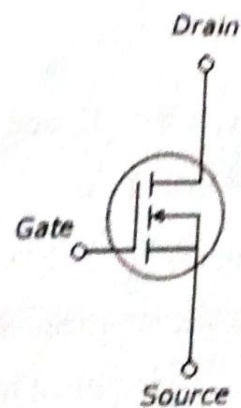
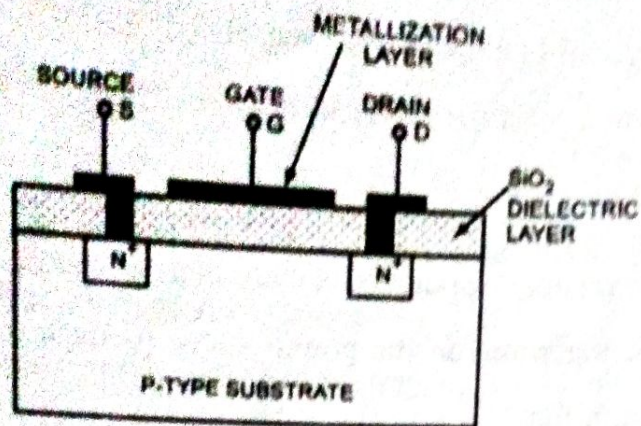
7. Discuss the structure and working of MOSFET.(May-2016)

MOSFET

The MOSFET is an abbreviation of metal oxide semiconductor field effect transistor. In MOSFET, the gate is insulated from the channel by using SiO_2 layer. The input impedance of MOSFET is high, because the gate current is very small. It is also called insulated Gate FET (IGFET). There are two types of MOSFETs. They are,

- i.) Enhancement MOSFET (E-MOSFET)
- ii.) Depletion MOSFET (DE- MOSFET)

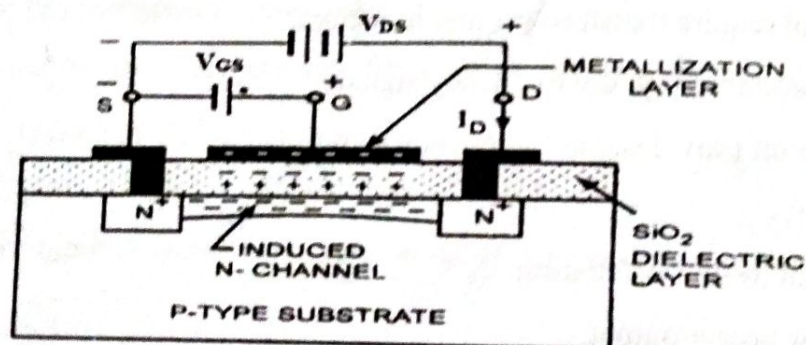
N-Channel Enhancement MOSFET



Construction & Symbol of N-Channel Enhancement MOSFET

Two heavily doped N type regions are diffused into a highly doped P type substrate. The two N type regions are source and drain. A thin layer of SiO_2 is grown over its surface. Then a metal contact is provided at the top of the SiO_2 layer, acts as gate. The oxide layer provides high input impedance to MOSFET.

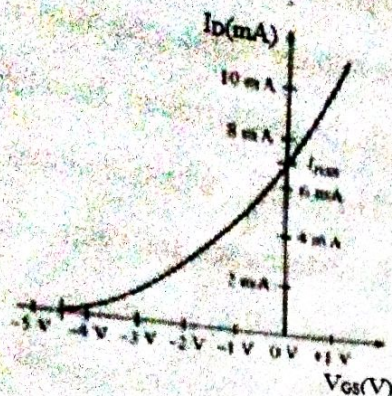
Operation of E-MOSFET:



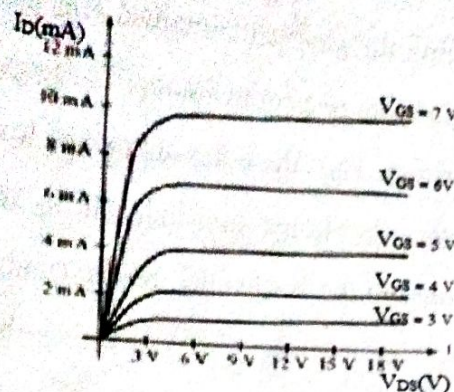
Operation of E-MOSFET

A proper biasing of N-channel E-MOSFET is given in fig., The gate is driven by only drain potential (V_{GS}) with respect to source. The drain is kept positive potential (V_{DS}) with respect to source. When V_{GS} is equal to zero, the PN junction between drain and substrate is reverse biased. Hence only a small leakage current (I_D) flows through the device.

When the gate is driven by positive potential, the electrons (minority carriers) in the P-substrate are moved towards the gate terminal. By the effect of layer SiO_2 (insulator), the electrons induce a N-channel between drain and source, which improve the conductivity. The minimum gate voltage, which produces N-channel layer between drain and source is called threshold voltage.



(a) Transfer Characteristics

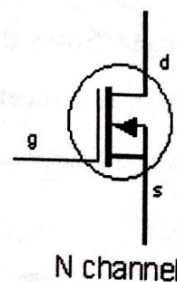
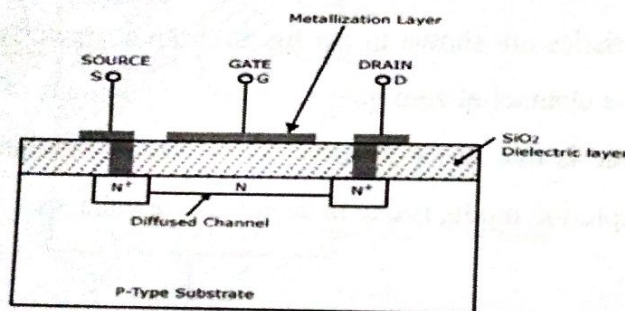


(b) Drain Characteristics

The drain and transfer characteristics of N-channel E-MOSFET are shown in the fig., From the figures we understand that the increasing gate voltage (V_{GS}) increase the drain current I_D . The amount of flow of drain current depends upon the width and density of the induced channel, which depends upon applied gate voltage.

In the transfer characteristics, the drain current is plotted against gate voltage (V_{GS}) at the constant levels of V_{DS} . At zero V_{GS} level, the current I_D is almost zero, because leakage current only flows. Then the drain current is increased linearly with the increasing of gate voltage.

N-channel depletion MOSFET:



N-channel depletion MOSFET with Symbol

The construction of an N-channel depletion MOSFET is shown in fig., It is similar as Enhancement MOSFET except that two highly doped N-regions are diffused into the P-Substrate. The two N-regions are source and drain. Here a lightly doped N-channel is diffused in between the regions source and drain. A thin layer of SiO_2 is grown over its surface. Then an aluminium layer is formed over on SiO_2 , acts as gate.

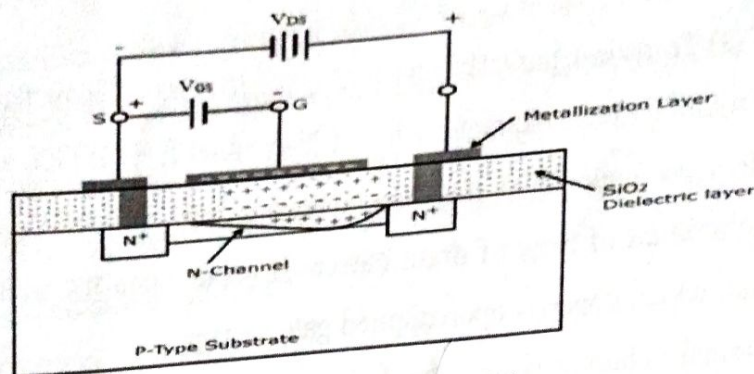
Operation:

The depletion MOSFET can be operated two modes. They are,

i.) Depletion mode

ii.) Enhancement mode

In depletion the gate is maintained at negative potential and the drain is maintained at positive potential with respect to source. The negative voltage on the gate attracts the holes in the P-type substrate. Thus the holes are moved towards the gate terminal. Because of the SiO_2 (insulator) layer, the holes are induced in the N-channel, which reduces the charge carriers (electrons) in the N-channel. So the conductivity decreases, the value of drain current also decreases.

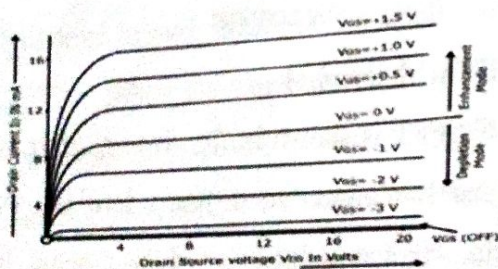


Depletion Mode Operation

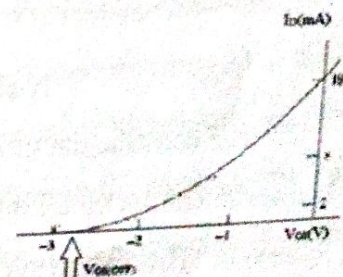
In enhancement mode both the gate and drain are maintained at positive potential with respect to source. The positive gate voltage induces the charges in the N-Channel. Hence the conductivity increases and the value of drain current is also increases.

Drain and transfer characteristics:

The drain and transfer characteristics are shown in the fig. in drain characteristics a valuable drain current flows through the channel at zero gate potential. The amount of drain current increasing in enhancement mode is due to the positive gate voltages. Similarly the amount of drain current decreases in depletion mode, is due to negative gate voltages.



(a) Drain Characteristics



(b) Transfer Characteristics

The transfer characteristic represents the flow of drain current with a variation of V_{GS} , at the constant of V_{DS} potential. The drain current increases in enhancement mode and decreases in depletion mode.

Part-A

1. What is the basic difference between amplifier and oscillator? (May-2019)

- An amplifier is a circuit which amplifies the weak signal and raises the amplitude of the signal.
- An oscillator is a circuit which generates the AC waveforms of particular frequency for providing source to an electronic circuit.

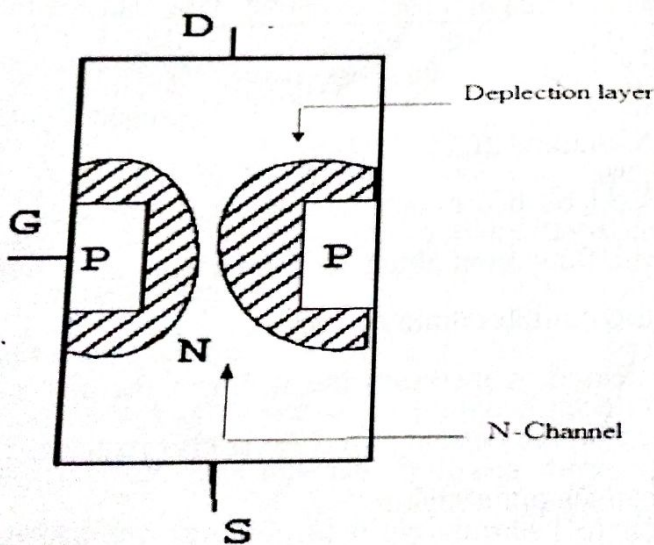
2. State any two functions of PN Junction diode.(May-2019)

The most common function of a diode is to allow an electric current to pass in one direction (called the diode's forward direction), while blocking it in the opposite direction (the reverse direction).

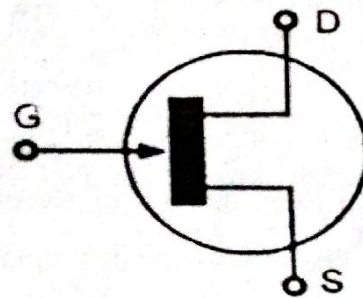
Part-B

1. Explain the construction and working of JFET with a neat diagram and necessary waveforms.(May-2019)

Junction Field Effect Transistor



Construction of N-channel JFET



Symbol of N-channel JFET

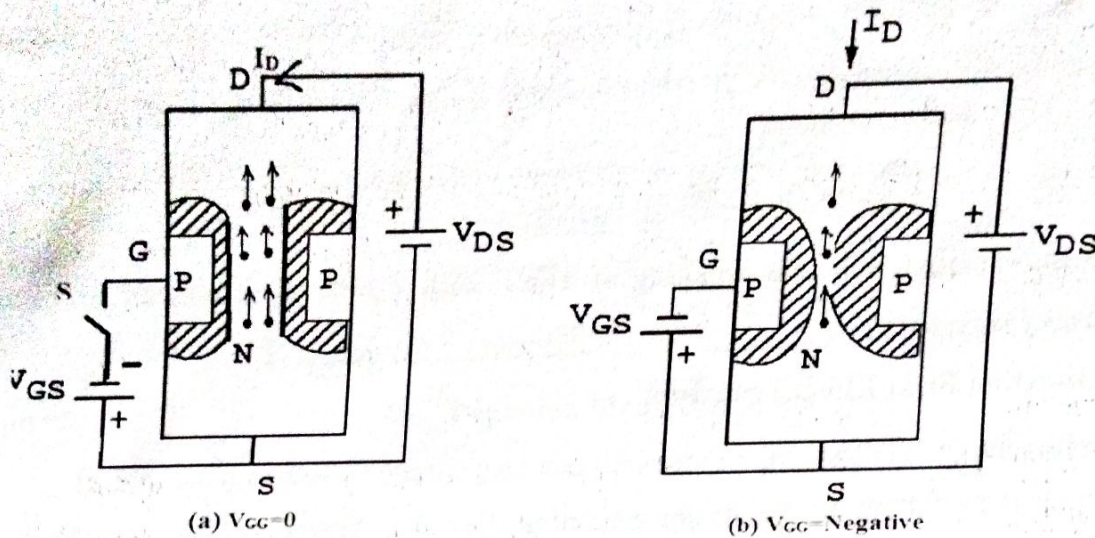
The construction of a N-channel JFET is shown in fig., it consists of an uniformly doped N-type semiconductor bar. On both sides of the semiconductor bar, two heavily doped P-type regions are formed by diffusion. The two P-regions are internally connected, and a single lead is taken out. This terminal is called "GATE". Ohmic contacts are made at the two ends of the N-type semi-conductor bar. One terminal is called "SOURCE" and the other is called "DRAIN".

The source is a terminal, where majority carriers enter the bar. The drain is a terminal where majority carriers leave the bar. The region between source and drain is called channel.

The majority carriers move from source to drain through this channel. The gate terminal controls the flow of majority carriers from source to drain.

Working principle of JFET

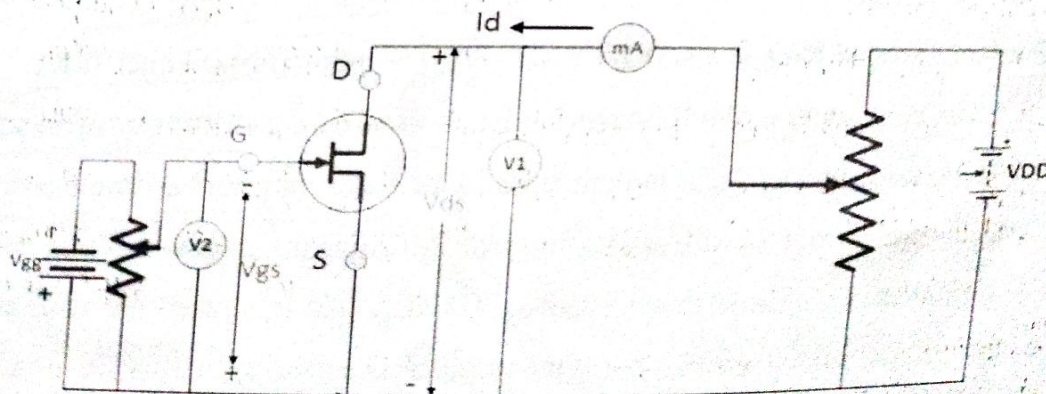
The N-channel JFET is normally biased by applying negative potential (V_{GS}) to the gate with respect to source. A positive potential (V_{DS}) is applied to the drain terminal with respect to source. The drain characteristics are obtained by taking the readings in between V_{DS} and I_D , with a constant variation of V_{GS} .



Operation of N-channel JFET

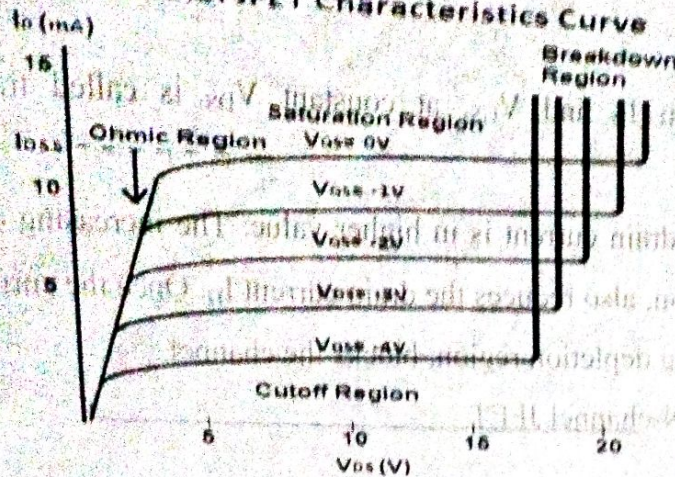
When $V_{GS}=0$, the two P-N junctions established a very thin and uniform depletion layer. Thus a large amount of electrons will flow from drain to source through a wide channel, in between the depletion region. This constitutes drain current I_D .

When the reverse voltage V_{GS} is increased, it increases the width of the depletion region. This reduces the width of the channel. So the current (I_D) flow is also reduced. The channel width is larger in the source region than the drain region.



Circuit diagram to determine the characteristics of JFET

N-Channel JFET Characteristics Curve



Drain characteristics of N-Channel JFET

Thus the current flowing through the channel is controlled by the reverse potential applied to the gate terminal. The circuit diagram to determine the characteristics of N-channel JFET is shown in fig.,

This circuit diagram is used to find out the drain and source characteristics of the JFET. The graph drawn between V_{DS} and I_D is known as drain characteristics or output characteristics. Three regions are formed in the drain characteristics; they are linear region, saturation region and breakdown region.

Linear region:

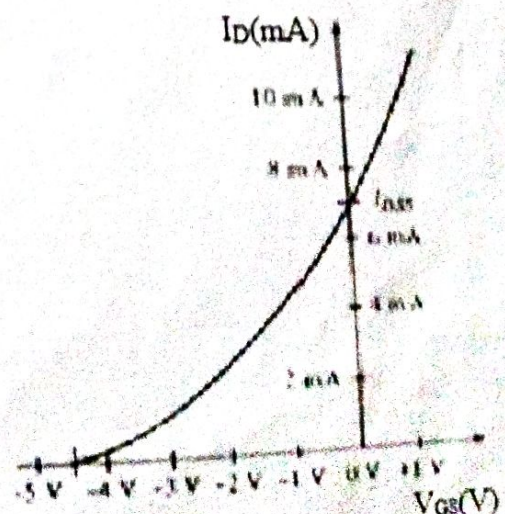
Initially as the drain to source voltage (V_{DS}) is increased, drain current (I_D) also increases. Such that I_D is directly proportional to V_{GS} . This region is called linear region or channel ohmic region.

Saturation region:

After a certain drain to source voltage (V_{DS}), the drain current (I_D) becomes almost constant. Now the channel becomes very narrow. Therefore, the variation of drain current (I_D) with the increasing of V_{DS} is constant. This region is called saturation region or pinch-off region. The minimum voltage, that the JFET produces the current as constant is called pinch-off voltage.

Breakdown region:

After a certain increase of V_{DS} , the drain current increases to a very high value, with a slight increase of V_{DS} . Now the depletion region at the PN junction undergo avalanche breakdown. This region is known as



breakdown region.

The graph between I_D and V_{GS} , at constant V_{DS} is called transfer or mutual characteristics.

When $V_{GS} = 0$, the drain current is in higher value. The increasing of V_{GS} , increases the width of depletion region, also reduces the drain current I_D . Once the current I_D is equal to zero, because the increasing depletion region, blocks the channel.

Transfer characteristics of N-channel JFET

From the graph,

- i.) Drain resistance (r_d) = $\frac{\Delta V_{DS}}{\Delta I_D}$ (at constant V_{GS})
- ii.) Transconductance (g_m) = $\frac{\Delta I_D}{\Delta V_{GS}}$ (at constant V_{DS})
- iii.) Amplification factor (μ) = $\frac{\Delta V_{DS}}{\Delta V_{GS}}$ (at constant I_D)

The relationship between μ , r_p and g_m is mentioned as.

$$\mu = r_d \times g_m$$

Applications:

- It is used as a buffer in measuring instruments.
- It is used in RF amplifiers of FM tuners and communications equipments.
- It is used in mixer circuits of FM and TV receivers.
- It is used in oscillator circuits.